Defend a Probability Claim

Week 10 Thursday — Show & Tell

COMSC 2043 • Discrete Structures and Critical Thinking

October 22, 2026

Today's destination

- Swap one denominator check.
- Defend a claim in four beats: claim, table or simulation, assumption, boundary.
- Ask and receive professional critique of the reasoning, not the reasoner.
- Revise your conclusion so it is no stronger than the evidence.

Warm-up: check a classmate's denominator

Scenario

1,000 people. Prevalence 2%. Sensitivity 80%. Specificity 90%.
A classmate reports: "$P(C \mid +) = 16/20 = 80\%$."

- Swap with a partner. Is the denominator right for $P(C \mid +)$?
- Rebuild the four-cell table, then confirm or correct.

The correct check

	Tests +	Tests −	Total
Has C (2%)	16	4	20
Not C (98%)	98	882	980
Total	114	886	1,000

- $P(C \mid +)$: denominator is the + column, 114. So $16/114 = 8/57 \approx 14.0\%$.
- $16/20 = 80\%$ answers a different question: $P(+ \mid C)$.

What Show & Tell is

- You make a claim public and defend it under questions.
- Not presentation theater: a hand calculation is enough.
- The evidence is the point. Polished slides earn nothing extra.
- You also give at least one specific, professional critique to a peer.

The critique norm

Critique the claim, assumption, evidence, inference, or model.

Do not attack the reasoner.

- Agreeing is not an exemption: probe whether the evidence deserves the confidence claimed.
- Revising a claim under challenge is not failure.
- Refusing to revise when the evidence demands it is not strength.

Questions that expose something

- What is the denominator, and why that one?
- Which assumption, if false, breaks this number? (representative population, independence, randomness)
- What did the simulation or second calculation check, and what could it never check?
- Is the conclusion stronger than the table supports?
- Where did the AI help, and how did you check it independently?

Your defense: four beats

1. **Claim** — one sentence.
2. **Table or simulation** — show the counts or the code and the check.
3. **Assumption** — name the one that matters most.
4. **Boundary** — what your evidence does not justify.

Then take questions. Keep the artifact inspectable: a hand-drawn table is fine.

Rotating defenses

Defend → question → rotate.

Take notes on the strongest challenge you receive and the critique you give.

Capture as you go

Strongest challenge received. What survived or changed. One critique you gave and why it was useful.

Peer questions and professional critique

- Ask the question you would want asked about your own claim.
- Name the thing you are challenging: claim, assumption, evidence, inference, or model.
- Say why it matters, then listen to the answer.

Not the same as an attack

"Your 90% is sensitivity, not $P(C \mid +)$" targets the inference. "You never get this stuff" targets the person.

Revise: make the conclusion no stronger than the evidence

Before

"A positive result means you probably have the condition."

After

"In the 10,000-person example with 1% prevalence, about 15 of every 100 positives have the condition (90/585). This holds only if the stated rates apply to the population."

The revision names the base rate, the number, and the boundary.

Work time: revise your conclusion and boundary

- Rewrite your one-sentence conclusion after the challenges you heard.
- State what survived, what you narrowed, and what remains unresolved.
- Note where AI helped and how you independently checked it.

Where this goes

These revisions feed your Decision Gate: Sources, Rules/Assumptions, Work, Check, One-Sentence Summary.

Show & Tell and the Decision Gate are different things

- **Show & Tell** is today's in-class practice: defend, question, critique, revise. No separate Canvas assignment.
- **Decision Gate** is the one submission this week: an individual write-up with the five headings.
- Do not submit the same artifact twice, and do not write an extra essay.

The due date and time are on the Decision Gate in Canvas.

Before you leave: your submission plan

1. Which claim will your Decision Gate defend? (screening example, a stated game or drawing process, or an approved bounded claim)
2. What is your Check, and what can it not validate?
3. Which of today's critiques changed your Rules/Assumptions?
4. When will you draft the one-sentence summary?

Next week: relations

- From probability over events to relations over pairs.
- Which pairs are related, and how do we classify the relation?
- Keep your Week 10 trace: it feeds the Decision Gate.

Exit question

What is one claim you narrowed today, and what did the evidence not justify?